

HEENAL PATEL

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EDUCATION

Sardar Patel Institute of Technology (SPIT), Mumbai

B.Tech. in Computer Engineering

Nov. 2022 - June 2026

CGPA: 8.28/10.0

Relevant Coursework:

Distributed Computing, Data Structures, Algorithms, Database Management Systems, Operating Systems, Computer Networks, OOPs, Artificial Intelligence & Machine Learning, System Programming, Cloud Computing, Generative AI Fundamentals, Building LLMs (GPTs)

RELEVANT EXPERIENCE

Tech Intern – LMS Development

North Konkan Chamber of Commerce & Agriculture (NKCCA)

Mar 2025 - Present

Mumbai, Maharashtra

- Developing an integrated **Learning Management System (LMS)** to support education initiatives across North Konkan region and architected a **scalable real-time system** with Firebase, applying robust **system design patterns** across platforms.
- Leading the build of a **Flutter-based tablet app** and a **React web portal** with features like **attendance tracking, homework uploads, study material access**, and **test monitoring** for students, parents, and admins.
- Integrated **Firestore, and Cloud Storage** for secure user management and real-time data handling .

Research Intern [Frontend and UI/UX Developer]

Aim4U

Jan. 2025 - Present

Mumbai, Maharashtra

- Crafting **intuitive and user-friendly digital experiences** for projects like **AI4Climate, KEM Physiotherapy, and AutoBuddys.in**, ensuring **seamless interaction and accessibility**.
- Building a **React-based web platform** where an **ML model scans plant images**, identifies them, and provides **medicinal benefits**, while also displaying **skincare products** made from the plant for purchase.
- Ensuring seamless **AWS integration** for hosting and scalability.

UI/UX Designer and App Developer Intern

Student Monitoring App

Feb. 2025 - Present

Mumbai, Maharashtra

- Designed and developed an **AI-powered student monitoring application**, integrating **ML-based frame presence detection** to ensure study adherence and accountability.
- Created the **complete UI/UX**, including **Figma layouts, interactive screen flows**, and user experience strategies.
- Built the **Flutter-based frontend**, ensuring a **responsive, engaging, and visually appealing** interface.
- Currently in **MVP Phase 1 testing with 100+ users**, optimizing overall performance and engagement.

ACHIEVEMENTS

10x Hackathon Winner | Including:

Smart India Hackathon 2024 | National Winner (315 winning teams from **500,000 participants**)

Dec. 2024

Other Notable Wins:

1st Place – K! Hacks 2.0 (**Anna Univ**, 1200+ teams), Hacknovare (**IEST Shibpur**, 100+), Hackaurora (**VJTI**, 100+)

2nd Place – M# (**MIT Manipal**, 600+), Odoo x AML (**IIT Gandhinagar**, 100+), Lean In Hacks (**IGDTUW**, 200+)

3rd Place – **HERE Tech Hack**, Smart City Hack (**IIT Guwahati**, 350+), **Track Win** at Quasar 3.0 (**VPPCOE**, 700+)

Co-Inventor of a **published patent** titled “PETHUB - An Integrated System for Comprehensive Pet Care”, published by **the Office of the Controller General of Patents, India** (Application No. 202421102429, Published on 31/01/2025)

NOTABLE PROJECTS

Dhara | React.js, Python, Node.js, Flutter, Firebase

Dec. 2024

- Developed a comprehensive **real-time safety management platform** for coal mines, integrating technologies to enhance **operational efficiency and worker safety** using **app for shift incharges and website for mine managers**.
- Built a **ML-powered risk management system** featuring **geospatial mapping, hazard response, predictive analytics** for production forecasting, maintenance scheduling, and **automated emergency triggers**, featuring **5 ML productivity enhancing tools** and support for **50+ languages**.
- Engineered an **advanced emergency response framework** integrating **ML-based document processing, automated shift log summaries, swarm evacuation protocols**, and **real-time communication tools**, tested with **2 IoT sensors** for **4 potential hazards**.

HealthWise | Flutter, Firebase, Flask, TensorFlow, BLE, WatchOS, Dart

Feb. 2025

- Engineered a **multi-platform Remote Patient Monitoring System (RPMS)** deployed across 3 apps (Patient, Doctor, WatchOS), enabling **real-time health tracking, automated SOS alerts**, and **smart vitals anomaly detection** for 24/7 proactive care.
- Integrated **50+ production-grade features** including **QR-based Smart Health ID, symptom checker, telemedicine suite**, and **doctor-patient collaboration tools**, supporting **5+ vitals parameters, 6+ AI models**, and **BLE-based wearable syncing**.
- Architected the backend for **high-throughput vitals ingestion** (tested with **50+ simulated users/sec**) using **Firestore sharding, BLE sync**, and **Cloud Functions** for real-time alerting.

TECHNICAL SKILLS

Web Development: HTML, CSS, JavaScript, React.js, Node.js, Material-UI, Git, npm, VS Code, WordPress

Database Systems: MySQL, Firebase, MongoDB

UI/UX: Figma, Adobe Illustrator, Wireframing, Prototyping, Responsive Design, User Research

Mobile Development: Flutter, Provider

CS Fundamentals: DSA, OOPs, System Design , Computer Architecture, Networks, DBMS, OS

Programming Languages: OOP (Java, C++, Dart), Scripting (Python, JavaScript/Node.js), Systems (C)